

Task 1

Task Name: Create a Car class using constructor and display method.

Objective:

- To learn how to create a class in Java.
- To understand constructor method.
- To learn how to initialize object using constructor.
- To practice method creation.
- To display object information using a method.

Problem Analysis:

In this problem, we need to create a class named Car. The class will contain some data members like brand, model and price. A constructor will be used to initialize these values. Then a method named displayCar() will print the car information.

Algorithm:

Step 1: Start

Step 2: Create a class named Car

Step 3: Declare variables (brand, model, price)

Step 4: Create constructor to initialize values

Step 5: Create displayCar() method

Step 6: Create object in main method

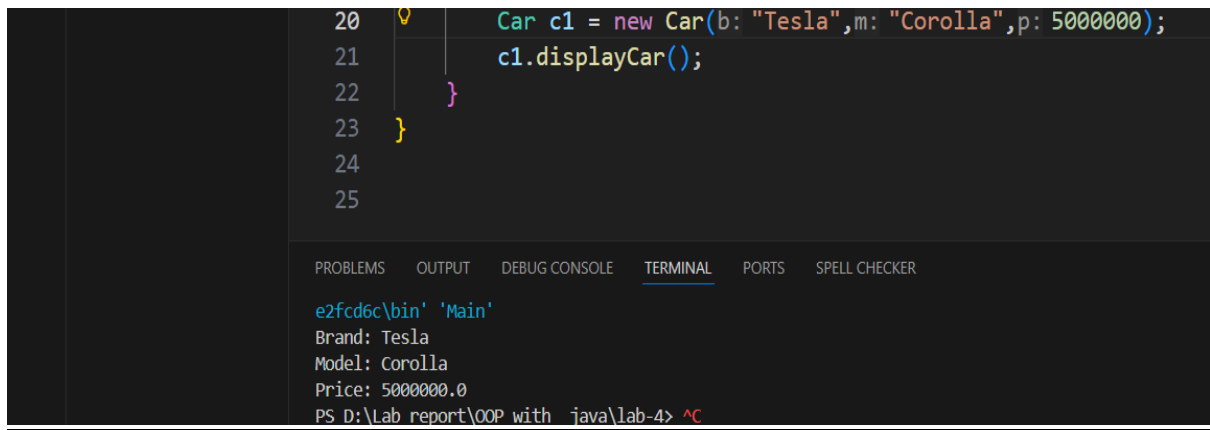
Step 7: Call displayCar() method

Step 8: End

Source Code:

<pre>class Car { String brand; String model; double price; Car(String b, String m, double p) { brand = b; model = m; price = p; } void displayCar() { System.out.println("Brand: " + brand); System.out.println("Model: " + model); System.out.println("Price: " + price); } }</pre>	<pre>} public class Main { public static void main(String[] args) { Car c1 = new Car("Toyota","Corolla",2000000); c1.displayCar(); } }</pre>
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Output:



The screenshot shows an IDE with a Java file. The code defines a `Car` class with a constructor and a `displayCar()` method. The constructor initializes `brand` to "Tesla", `model` to "Corolla", and `price` to 5000000.0. The `displayCar()` method prints the object's details. The output window shows the execution of the `main` method, displaying the car's brand, model, and price.

```
20     Car c1 = new Car(b: "Tesla",m: "Corolla",p: 5000000);
21     c1.displayCar();
22 }
23 }
24
25
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER

e2fcd6c\bin' 'Main'
Brand: Tesla
Model: Corolla
Price: 5000000.0
PS D:\Lab report\OOP with java\lab-4> ^C

Discussion & Conclusion:

This program demonstrates the basic concepts of Object-Oriented Programming (OOP) in Java by creating a `Car` class using a constructor and a display method. The constructor is used to initialize the values of the object such as brand, model, and price during object creation, while the display method is used to print the object information. Through this experiment, we learned how to create classes and objects, how constructors work, and how methods are used to display data in Java. We also gained practical knowledge about object initialization, data handling, and code organization in OOP. The program was executed successfully and produced the expected output, which helped in understanding the importance of constructors and methods in Java programming.

Task 2

Task Name: Create an IDCard class using constructor and show method.

Objective:

- To understand class and object concepts.
- To learn constructor initialization.
- To store student information.
- To create user defined methods.
- To display object data.

Problem Analysis:

Here we will create a class called IDCard. It will contain name, id, department and institution. A constructor will assign values when the object is created. A method showID() will display the information.

Algorithm:

Step 1: Start

Step 2: Create IDCard class

Step 3: Declare variables (name, id, department, institution)

Step 4: Create constructor

Step 5: Create showID() method

Step 6: Create object

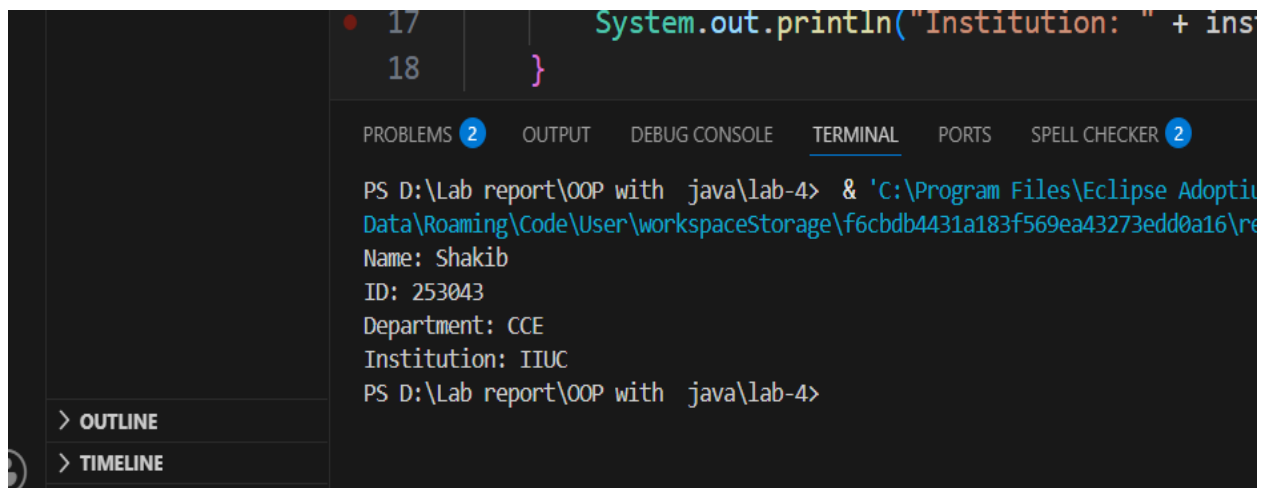
Step 7: Call showID()

Step 8: End

Source Code:

<pre>class IDCard { String name; int id; String department; String institution; IDCard(String n, int i, String d, String ins){ name = n; id = i; department = d; institution = ins; } void showID() { System.out.println("Name: " + name); System.out.println("ID: " + id);</pre>	<pre>System.out.println("Department: " + department); System.out.println("Institution: " + institution); } } public class Main { public static void main(String[] args) { IDCard s1 = new IDCard("Saif",22135,"CSE","ABC College"); s1.showID(); } }</pre>
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Output:



The screenshot shows the Eclipse IDE interface. At the top, a snippet of Java code is visible, showing lines 17 and 18. Line 17 contains `System.out.println("Institution: " + ins` and line 18 contains a closing curly brace `}`. Below the code editor, the 'TERMINAL' tab is active, displaying the output of a Java program. The output text is as follows:

```
PS D:\Lab report\OOP with java\lab-4> & 'C:\Program Files\Eclipse Adopti
Data\Roaming\Code\User\workspaceStorage\f6cbdb4431a183f569ea43273edd0a16\re
Name: Shakib
ID: 253043
Department: CCE
Institution: IIUC
PS D:\Lab report\OOP with java\lab-4>
```

On the left side of the IDE, there are buttons for '> OUTLINE' and '> TIMELINE'.

Discussion & Conclusion:

This program demonstrates the use of Object-Oriented Programming (OOP) concepts in Java by creating an IDCard class with a constructor and a show method. The constructor is used to initialize the information of the ID card such as name, ID number, and department at the time of object creation, while the show method displays the stored information. Through this experiment, we learned how to create classes and objects, how constructors are used for automatic data initialization, and how methods help in displaying object data efficiently. We also gained practical knowledge about organizing data in an object-oriented way and improving code readability and reusability. The program was executed successfully and produced the expected output, which helped us understand the basic implementation of constructors and methods in Java OOP.